

Throughout this course, I developed a much better understanding of how operating systems manage the interaction between hardware, software, and users. At the beginning of the course, I viewed an operating system mainly as the software responsible for running applications. As the course progressed, I learned that operating systems are also responsible for process scheduling, memory management, resource allocation, synchronization, file systems, and security. The combination of written assignments and programming projects helped me understand how these concepts work together to create reliable and efficient computer systems.

One of the biggest challenges I encountered was learning how to use Python libraries that interact directly with operating system resources. Using the psutil library to retrieve CPU, memory, disk, and process information required understanding both the library documentation and the operating system concepts behind the information being displayed. Later in the course, implementing multithreading and semaphores introduced another challenge because I had to understand how multiple tasks execute concurrently while safely sharing system resources. I overcame these challenges by reading the documentation carefully, testing the programs incrementally, and breaking larger problems into smaller pieces until I understood how each component worked.

The assignments gave me many opportunities to apply operating system principles and best practices. Throughout the coding exercises, I worked with process management, CPU monitoring, memory usage, threading, synchronization, and fault

handling. During Project One, I evaluated Windows Server 2025 and analyzed how operating system architecture, security, and resource management influence enterprise environments. Project Two allowed me to apply many of those concepts by developing an operating system performance simulator that monitored system resources while demonstrating memory management, scheduling, threading, and error handling.

The knowledge that will be most beneficial to my future career is understanding how operating systems manage resources and how those decisions affect software performance. As I continue working toward my master's degree in computer science with a concentration in artificial intelligence, this knowledge will help me build applications that are more efficient, scalable, and reliable. Understanding synchronization, memory management, and resource allocation will also be valuable when developing applications that process large amounts of data or execute multiple tasks simultaneously.

One aspect of the course that surprised me was how relevant operating system concepts are to software development. Before taking this course, I expected to learn mostly theoretical concepts. Instead, many assignments required implementing practical solutions using Python, making it easier to understand how these concepts are applied in real-world systems.

Overall, this course strengthened both my theoretical understanding and my practical programming skills. It provided a solid foundation in operating system concepts that I will continue applying throughout my future coursework, personal projects, and professional career.